

# Inequalities of degree greater than or equal to 3

MathE

27<sup>th</sup> of April of 2023

**Exercise 1.** Find the values of  $x$  that satisfy the inequality  $x^3 + 6x^2 - 6 > 2x^2 - x$ .

Step 1: Put all terms on the same side of the inequality.

$$\begin{aligned} x^3 + 6x^2 - 6 &> 2x^2 - x \iff \\ x^3 + 6x^2 - 6 - 2x^2 + x &> 2x^2 - x - 2x^2 + x \iff \\ x^3 + 4x^2 + x - 6 &> 0 \end{aligned}$$

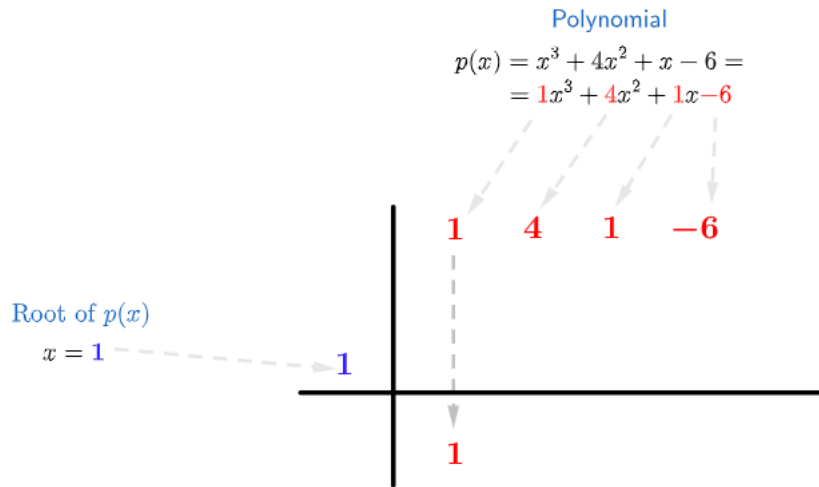
Step 2: Find a root of the polynomial  $p(x) = x^3 + 4x^2 + x - 6$  (usually, it is a good idea to try 0, 1, -1, 2 or -2).

For example, you can ask yourself:

- Is 0 a root of  $p(x)$ ?
  - The answer is no, because  $p(0) = 0^3 + 4 \cdot 0^2 + 0 - 6 = -6 \neq 0$ .
- Is -1 a root of  $p(x)$ ?
  - The answer is no, because  $p(-1) = (-1)^3 + 4 \cdot (-1)^2 + (-1) - 6 = -1 + 4 - 1 - 6 = -4 \neq 0$ .
- Is 1 a root of  $p(x)$ ?
  - The answer is yes, because  $p(1) = 1^3 + 4 \cdot 1^2 + 1 - 6 = 1 + 4 + 1 - 6 = 0$ .

So we found a root of the polynomial  $p(x)$ , which is  $x = 1$  (there can be other roots, but we only need one).

Step 3: Draw a table in the following way:

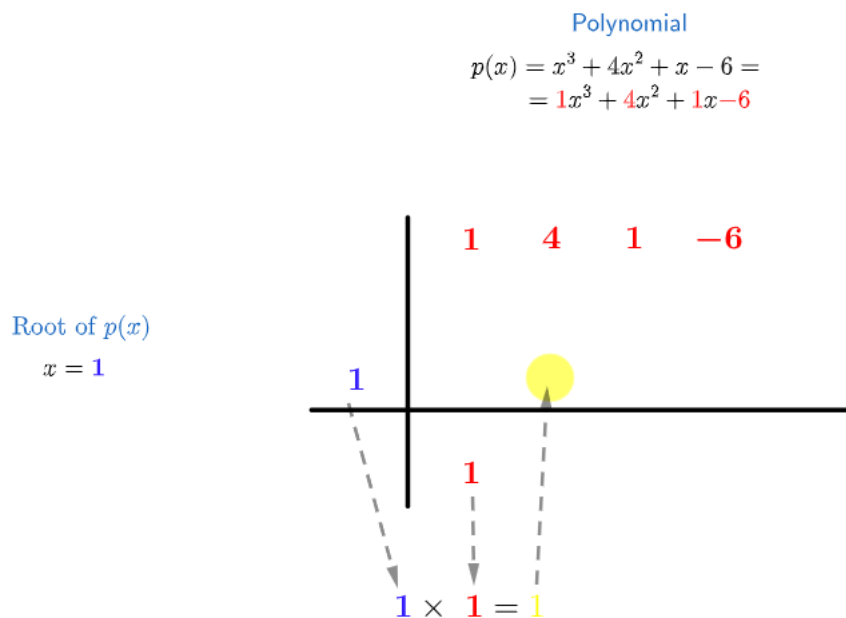


As we can see in the image above:

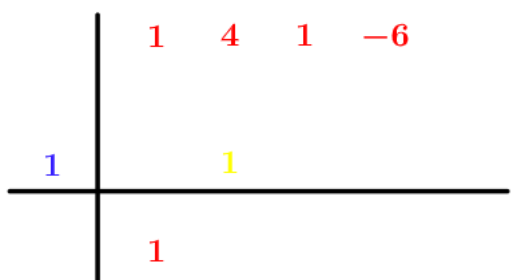
1. The coefficients of the polynomial are written in the first line respecting the degree of  $x$ ; that is, we start by writing the coefficients of the powers of  $x$  with the highest degree from left to right.
2. The root stays above the horizontal line segment and to the left of the vertical line segment.
3. The coefficient of  $x$  with the highest degree comes down.

(in this case it is 1 because  $x^3$  is the term of the polynomial with the highest degree).

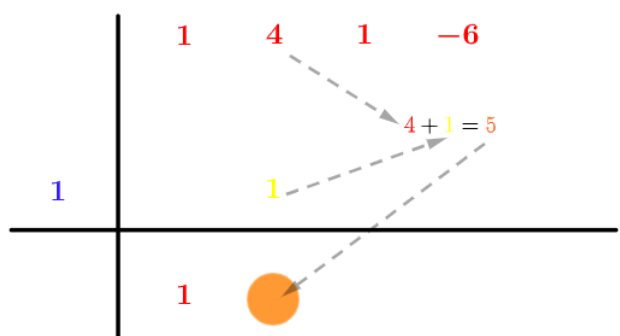
Step 4:



As a result, we get



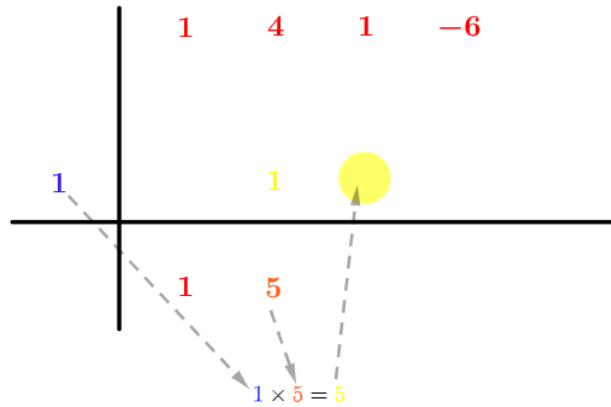
Step 5:



As a result, we get



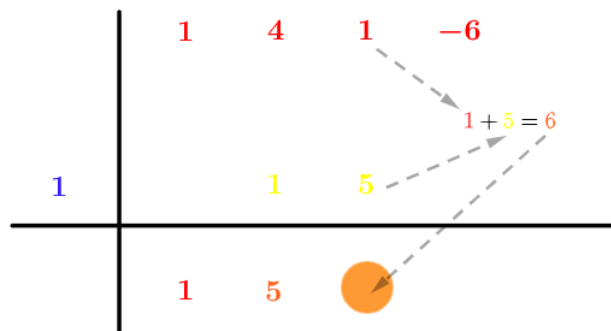
Step 6:



As a result, we get



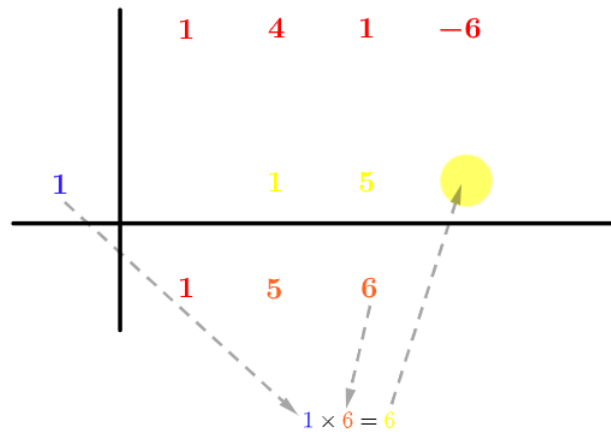
Step 7:



As a result, we get



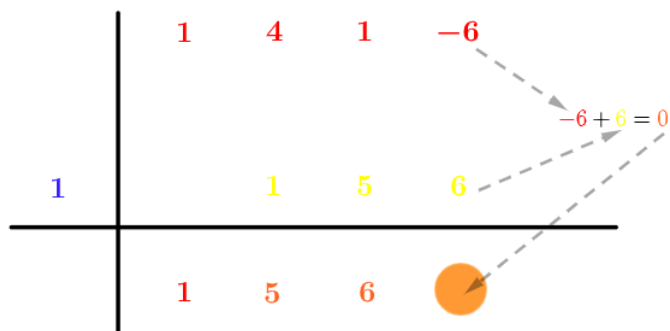
Step 8:



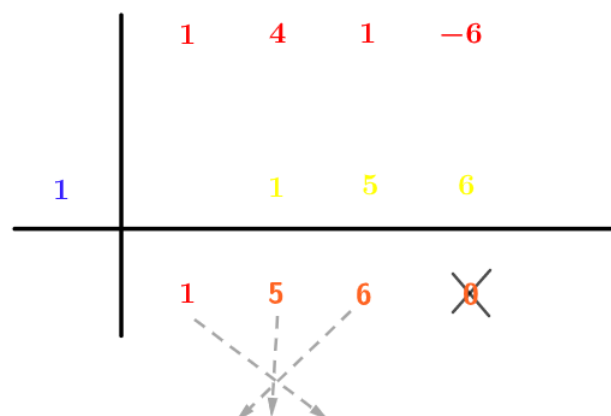
As a result, we get



Step 9:



As a result, we get



The numbers in *orange* give us the polynomial  $6 + 5x + 1x^2$  (we “ignore” the last number, which is always 0).

As a result, we get that

$$\underbrace{x^3 + 4x^2 + x - 6}_{p(x)} = (x - 1) \cdot (x^2 + 5x + 6)$$

$x - 1$  is the binomial obtained by subtracting the root found on Step 2 to  $x$ .

$x^2 + 5x + 6$  is the trinomial we got from Ruffini's Rule.

This is helpful because now solving  $p(x) = 0$  is equivalent to solve  $(x - 1) \cdot (x^2 + 5x + 6) = 0$ :

$$\begin{aligned}
 (x - 1) \cdot (x^2 + 5x + 6) &= 0 \iff \\
 x - 1 = 0 \vee x^2 + 5x + 6 &= 0 \iff \\
 x = 1 \vee x = \frac{-5 \pm \sqrt{5^2 - 4 \times 1 \times 6}}{2 \times 1} &\iff \\
 x = 1 \vee x = \frac{-5 \pm \sqrt{25 - 24}}{2} &\iff \\
 x = 1 \vee x = \frac{-5 \pm \sqrt{1}}{2} &\iff \\
 x = 1 \vee x = \frac{-5 \pm 1}{2} &\iff \\
 x = 1 \vee x = \frac{-5 - 1}{2} \vee x = \frac{-5 + 1}{2} &\iff \\
 x = 1 \vee x = -3 \vee x = -2 &
 \end{aligned}$$

We just found out that the roots of  $p(x)$  are  $-3, -2$  and  $1$ .

Because  $p(x)$  is a continuous function, it follows from the Bolzano's theorem that the sign of the function on the interval  $(-\infty, -3)$  will not change (it will either be positive or negative).

The same happens on the intervals  $(-3, -2)$ ,  $(-2, 1)$  and  $(1, +\infty)$ .

To see if the sign is either positive or negative, we can evaluate  $p(x)$  at any point of the interval:

- On the interval  $(-\infty, -3)$ ,  $p(x)$  is negative because  $-4 \in (-\infty, -3)$  and  $p(-4) = -10 < 0$ .
- On the interval  $(-3, -2)$ ,  $p(x)$  is positive because  $-2.5 \in (-3, -2)$  and  $p(-2.5) = 0.875 > 0$ .
- On the interval  $(-2, 1)$ ,  $p(x)$  is negative because  $0 \in (-2, 1)$  and  $p(0) = -6 < 0$ .
- On the interval  $(1, +\infty)$ ,  $p(x)$  is positive because  $2 \in (1, +\infty)$  and  $p(2) = 20 > 0$ .

As a result, we can construct the following table:

|        |           |     |      |     |      |     |     |     |           |
|--------|-----------|-----|------|-----|------|-----|-----|-----|-----------|
| $x$    | $-\infty$ |     | $-3$ |     | $-2$ |     | $1$ |     | $+\infty$ |
| $p(x)$ | $-$       | $-$ | $0$  | $+$ | $0$  | $-$ | $0$ | $+$ | $+$       |

We saw that the original inequality, which is

$$x^3 + 6x^2 - 6 > 2x^2 - x$$

is equivalent to the inequality

$$x^3 + 4x^2 + x - 6 > 0$$

The table above tells us that the values of  $x$  that satisfy this last inequality are

$$x \in (-3, -2) \text{ and } x \in (1, +\infty)$$

As a result, the solution is  $x \in (-3, -2) \cup (1, +\infty)$ .

**Answer:**  $x \in (-3, -2) \cup (1, +\infty)$

**Exercise 2.** Solve the inequality  $4x^4 < 3 - 11x^2$ .

Step 1: Put all terms on the same side of the inequality.

$$\begin{aligned}4x^4 &< 3 - 11x^2 \iff \\4x^4 - \textcolor{red}{3} + \textcolor{red}{11x^2} &< 3 - 11x^2 - \textcolor{red}{3} + \textcolor{red}{11x^2} \iff \\4x^4 + 11x^2 - 3 &< 0\end{aligned}$$

Step 2: Find a root of the polynomial  $q(x) = 4x^4 + 11x^2 - 3$ .

In order to do that, we first do the change of variable  $y = x^2$ :

$$\begin{aligned}q(x) &= 4x^4 + 11x^2 - 3 = \\4(x^2)^2 + 11x^2 - 3 &= \\4y^2 + 11y - 3\end{aligned}$$

We did that because we know how to solve a quadratic equation (we use the quadratic formula):

$$\begin{aligned}q(x) = 0 &\iff \\4y^2 + 11y - 3 &= 0 \iff \\y = \frac{-11 \pm \sqrt{11^2 - 4 \times 4 \times (-3)}}{2 \times 4} &\iff \\y = \frac{-11 \pm \sqrt{121 + 48}}{8} &\iff \\y = \frac{-11 \pm 13}{8} &\iff \\y = \frac{-11 - 13}{8} \vee y = \frac{-11 + 13}{8} &\iff \\y = \frac{-24}{8} \vee y = \frac{2}{8} &\iff \\y = -3 \vee y = \frac{1}{4}\end{aligned}$$

We now take into account the change of variable we did before:

$$\begin{aligned}y = x^2 &\iff \\ \underbrace{-3 = x^2}_{\textcolor{blue}{(1)} \text{ impossible condition in } \mathbb{R}} \vee \frac{1}{4} = x^2 &\iff \\\frac{1}{4} = x^2 &\iff \\x = \pm \sqrt{\frac{1}{4}} &\iff \\x = \pm \frac{1}{2}\end{aligned}$$

(1) We have an impossible condition because, in the real numbers, a number squared is always non-negative, so it cannot be equal to  $-3$ .

This tells us the real roots of the polynomial  $q(x) = 4x^4 + 11x^2 - 3$  are  $-\frac{1}{2}$  and  $\frac{1}{2}$ .

As in the previous exercise, because  $q(x)$  is a continuous function, it follows from the Bolzano's theorem that the sign of the function on the interval  $(-\infty, -\frac{1}{2})$  will not change (it will either be positive or negative).

The same happens on the intervals  $(-\frac{1}{2}, \frac{1}{2})$  and  $(\frac{1}{2}, +\infty)$ .

To see if the sign is either positive or negative, we can evaluate  $q(x)$  at any point of the interval:

- On the interval  $(-\infty, -\frac{1}{2})$ ,  $q(x)$  is positive because  $-1 \in (-\infty, -\frac{1}{2})$  and  $q(-1) = 11 > 0$ .
- On the interval  $(-\frac{1}{2}, \frac{1}{2})$ ,  $q(x)$  is negative because  $0 \in (-\frac{1}{2}, \frac{1}{2})$  and  $q(0) = -3 < 0$ .
- On the interval  $(\frac{1}{2}, +\infty)$ ,  $q(x)$  is positive because  $1 \in (\frac{1}{2}, +\infty)$  and  $q(1) = 11 > 0$ .

As a result, we can construct the following table:

|        |           |   |                |   |               |   |           |
|--------|-----------|---|----------------|---|---------------|---|-----------|
| $x$    | $-\infty$ |   | $-\frac{1}{2}$ |   | $\frac{1}{2}$ |   | $+\infty$ |
| $q(x)$ | +         | + | 0              | − | 0             | + | +         |

We saw that the original inequality, which is

$$4x^4 < 3 - 11x^2$$

is equivalent to the inequality

$$4x^4 + 11x^2 - 3 < 0$$

The table above tells us that the values of  $x$  that satisfy this last inequality are

$$x \in \left(-\frac{1}{2}, \frac{1}{2}\right)$$

As a result, the solution is  $x \in (-\frac{1}{2}, \frac{1}{2})$ .

**Answer:**  $x \in (-\frac{1}{2}, \frac{1}{2})$

**Exercise 3.** Solve the inequality  $x^4 - 25x^2 < -2x^3 + 26x - 120$ .

Step 1: Put all terms on the same side of the inequality.

$$\begin{aligned}x^4 - 25x^2 &< -2x^3 + 26x - 120 \iff \\x^4 - 25x^2 + 2x^3 - 26x + 120 &< -2x^3 + 26x - 120 + 2x^3 - 26x + 120 \iff \\x^4 + 2x^3 - 25x^2 - 26x + 120 &< 0\end{aligned}$$

Step 2: Find a root of the polynomial  $r(x) = x^4 + 2x^3 - 25x^2 - 26x + 120$  (usually, it is a good idea to try 0, 1, -1, 2 or -2).

For example, you can ask yourself:

- Is 0 a root of  $r(x)$ ?
  - The answer is no, because  $r(0) = 0^4 + 2 \times 0^3 - 25 \times 0^2 - 26 \times 0 + 120 = 120 \neq 0$ .
- Is 1 a root of  $r(x)$ ?
  - The answer is no, because  $r(1) = 1^4 + 2 \times 1^3 - 25 \times 1^2 - 26 \times 1 + 120 = 72 \neq 0$ .

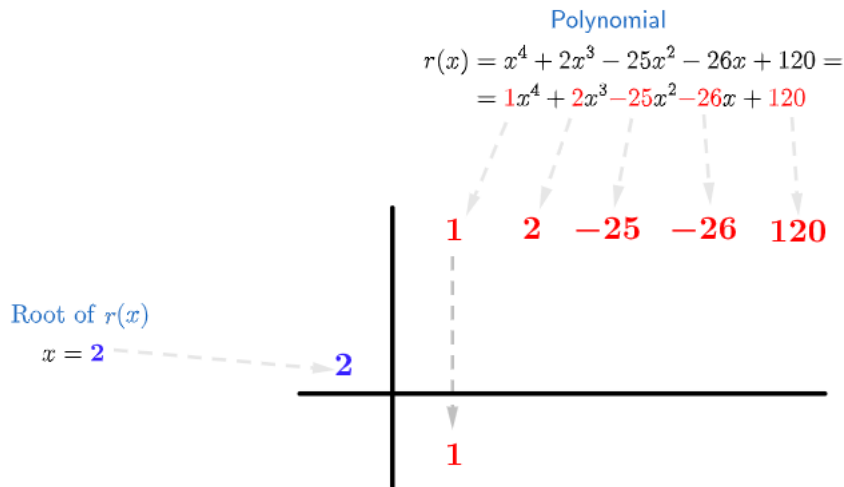
- Is -1 a root of  $r(x)$ ?
  - The answer is no, because

$$r(-1) = (-1)^4 + 2 \times (-1)^3 - 25 \times (-1)^2 - 26 \times (-1) + 120 = 120 \neq 0$$

- Is 2 a root of  $r(x)$ ?
  - The answer is yes, because  $r(2) = 2^4 + 2 \times 2^3 - 25 \times 2^2 - 26 \times 2 + 120 = 0$

So we found a root of the polynomial  $r(x)$ , which is  $x = 2$  (there can be other roots, but we only need one).

Step 3: Draw a table in the following way:



Proceeding as we did in Exercise 1, we obtain the following final table:

|   |   |   |     |      |              |
|---|---|---|-----|------|--------------|
|   | 1 | 2 | -25 | -26  | 120          |
| 2 | 2 | 8 | -34 | -120 |              |
|   | 1 | 4 | -17 | -60  | <del>0</del> |

The numbers in *orange* give us the polynomial  $-60 - 17x + 4x^2 + 1x^3$  (we “ignore” the last number, which is always 0).

As a result, we get that

$$x^4 + 2x^3 - 25x^2 - 26x + 120 = (x - 2) \cdot (x^3 + 4x^2 - 17x - 60)$$

This is helpful because now solving  $r(x) = 0$  is equivalent to solve  $(x - 2) \cdot (x^3 + 4x^2 - 17x - 60) = 0$ :

$$(x - 2) \cdot (x^3 + 4x^2 - 17x - 60) = 0 \iff$$

$$x - 2 = 0 \vee x^3 + 4x^2 - 17x - 60 = 0$$

$$\iff x = 2 \vee x^3 + 4x^2 - 17x - 60 = 0$$

The next step is to find the roots of  $x^3 + 4x^2 - 17x - 60 = 0$ .

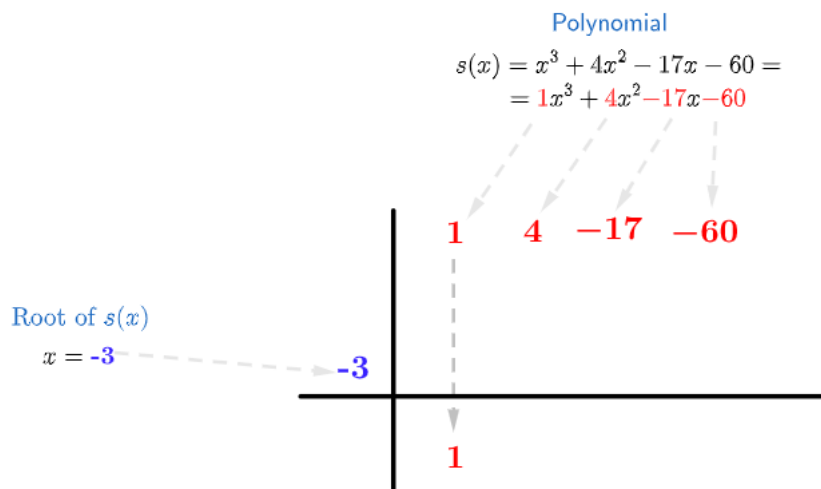
Step 4: Find a root of the polynomial  $s(x) = x^3 + 4x^2 - 17x - 60$  (usually, it is a good idea to try 0, 1, -1, 2 or -2).

For example, you can ask yourself:

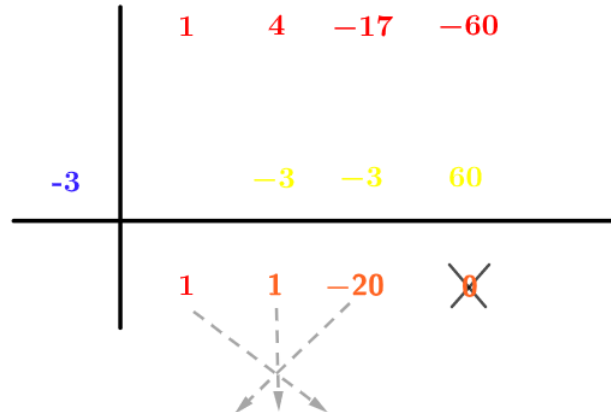
- Is 0 a root of  $s(x)$ ?
  - The answer is no, because  $s(0) = 0^3 + 4 \times 0^2 - 17 \times 0 - 60 = -60 \neq 0$ .
- Is 1 a root of  $s(x)$ ?
  - The answer is no, because  $s(1) = 1^3 + 4 \times 1^2 - 17 \times 1 - 60 = -72 \neq 0$ .
- Is -1 a root of  $s(x)$ ?
  - The answer is no, because  $s(-1) = (-1)^3 + 4 \times (-1)^2 - 17 \times (-1) - 60 = -40 \neq 0$
- (...)
- Is -3 a root of  $s(x)$ ?
  - The answer is yes, because  $s(-3) = (-3)^3 + 4 \times (-3)^2 - 17 \times (-3) - 60 = 0$

So we found a root of the polynomial  $s(x)$ , which is  $x = -3$  (there can be other roots, but we only need one).

Step 5: Draw a table in the following way:



Proceeding as we did before, we obtain the following final table:



The numbers in *orange* give us the polynomial  $-20 + 1x + 1x^2$  (we “ignore” the last number, which is always 0).

As a result, we get that

$$x^3 + 4x^2 - 17x - 60 = (x - (-3)) \cdot (x^2 + x - 20) = (x + 3) \cdot (x^2 + x - 20)$$

This is helpful because now solving  $s(x) = 0$  is equivalent to solve  $(x + 3) \cdot (x^2 + x - 20) = 0$ :

$$\begin{aligned} (x + 3) \cdot (x^2 + x - 20) &= 0 \iff \\ x + 3 &= 0 \vee x^2 + x - 20 = 0 \iff \\ x = -3 \vee x &= \frac{-1 \pm \sqrt{1^2 - 4 \times 1 \times (-20)}}{2 \times 1} \iff \\ x = -3 \vee x &= \frac{-1 \pm \sqrt{1 + 80}}{2} \iff \\ x = -3 \vee x &= \frac{-1 \pm \sqrt{81}}{2} \iff \\ x = -3 \vee x &= \frac{-1 \pm 9}{2} \iff \\ x = -3 \vee x &= \frac{-1 - 9}{2} \vee x = \frac{-1 + 9}{2} \iff \\ x = -3 \vee x &= \frac{-10}{2} \vee x = \frac{8}{2} \iff \\ x = -3 \vee x &= -5 \vee x = 4 \end{aligned}$$

We just found out that the roots of  $s(x)$  are  $-5, -3$  and  $4$ .

We saw the roots of  $r(x)$  are  $2$  and the roots of  $s(x)$ , so we conclude the roots of  $r(x)$  are  $-5, -3, 2$  and  $4$ .

Because  $r(x)$  is a continuous function, it follows from the Bolzano's theorem that the sign of the function on the interval  $(-\infty, -5)$  will not change (it will either be positive or negative).

The same happens on the intervals  $(-5, -3), (-3, 2), (2, 4)$  and  $(4, +\infty)$ .

To see if the sign is either positive or negative, we can evaluate  $r(x)$  at any point of the interval:

- On the interval  $(-\infty, -5)$ ,  $r(x)$  is positive because  $-6 \in (-\infty, -5)$  and  $r(-6) = 240 > 0$ .
- On the interval  $(-5, -3)$ ,  $r(x)$  is negative because  $-4 \in (-5, -3)$  and  $r(-4) = -48 < 0$ .
- On the interval  $(-3, 2)$ ,  $r(x)$  is positive because  $0 \in (-3, 2)$  and  $r(0) = 120 > 0$ .
- On the interval  $(2, 4)$ ,  $r(x)$  is negative because  $3 \in (2, 4)$  and  $r(3) = -48 < 0$ .
- On the interval  $(4, +\infty)$ ,  $r(x)$  is positive because  $5 \in (4, +\infty)$  and  $r(5) = 240 > 0$ .

As a result, we can construct the following table:

|        |           |   |      |   |      |   |     |   |     |   |           |
|--------|-----------|---|------|---|------|---|-----|---|-----|---|-----------|
| $x$    | $-\infty$ |   | $-5$ |   | $-3$ |   | $2$ |   | $4$ |   | $+\infty$ |
| $r(x)$ | +         | + | 0    | − | 0    | + | 0   | − | 0   | + | +         |

We saw that the original inequality, which is

$$x^4 - 25x^2 < -2x^3 + 26x - 120$$

is equivalent to the inequality

$$x^4 + 2x^3 - 25x^2 - 26x + 120 < 0$$

The table above tells us that the values of  $x$  that satisfy this last inequality are

$$x \in (-5, -3) \text{ and } x \in (2, 4)$$

As a result, the solution is  $x \in (-5, -3) \cup (2, 4)$ .

**Answer:**  $x \in (-5, -3) \cup (2, 4)$